Browser-Based CSV Duplicate Matcher

User Guide & Operational Documentation | Version 2.0

1. Executive Overview & Design Philosophy

The **CSV Duplicate Matcher & VLOOKUP Tool** is a single-file, browser-native utility designed to compare two datasets (a *Master List* and a *Secondary List*), detect overlapping records using hierarchical matching rules, and generate clean, segregated outputs.

Core Principles & Data Integrity Guarantee

- **Strict Structural Preservation:** The tool never alters, reorders, or removes column headers, formatting, or supplementary fields from the Master List or Secondary List.
- **100% Client-Side Processing:** All parsing and reconciliation occurs inside the user's browser memory using JavaScript and Web APIs. Zero data is transmitted to an external server.
- **Multi-Output Categorization:** Exports clean Master records, removed duplicate matches, and unmatched secondary entries into isolated CSV files.

2. System Outputs & CSV Exports

Upon running the deduplication process, the tool segregates your records into three distinct functional outputs:

Output 1

CLEANED MASTER LIST

Original Master CSV with all detected duplicate rows fully removed. All original columns remain unchanged.

Output 2

REMOVED MASTER MATCHES

A dedicated CSV containing all Master records that matched the Secondary List for audit trail purposes.

Output 3

UNMATCHED SECONDARY

Secondary list records that had no matching entry in the Master List (flagged missing leads/contacts).

3. Matching Engine Logic & Fallback Hierarchy

The matcher uses a two-tiered waterfall reconciliation engine. It evaluates incoming rows sequentially to ensure high precision while catching individuals who used varying contact details.

[Priority 1 - Exact Email Match]: Compare sanitized email address.

Normalizes strings by removing leading/trailing whitespace and converting all letters to lowercase (e.g., `John.Doe@Domain.com` → `john.doe@domain.com`).

[Priority 2 - Name Combination Fallback]: First Name + Last Name match.

If email is missing, blank, or fails to match, the tool concatenates normalized First Name and Last Name into a single key (e.g., `john|doe`) and evaluates secondary entries against this key.

4. Step-by-Step Operational Workflow

Step	Action Required	Expected Result
1 **Upload Files**	Select your Master CSV and Secondary CSV using the file picker cards.	Files are parsed instantaneously into client memory via PapaParse.
2 **Map Columns**	Verify or set dropdown mappings for Email, First Name, and Last Name.	Auto-detection maps matching header names automatically.
3 **Execute Matching**	Click the primary blue button *"Find & Remove Matching Records"*.	The engine processes lists and opens the Interactive Modal.
4 **Review Preview**	Use modal tabs to inspect matched records and unmatched secondary rows.	Display summary metrics and a preview of the first 50 matched rows.
5 **Export Files**	Click the designated green, blue, or orange buttons to download desired CSVs.	Browser triggers direct CSV download without server interaction.

Important Technical Notes

- **Header Requirements:** Files must be formatted as standard comma-separated CSVs with headers in the first row.
- **Memory Capacity:** Because processing occurs entirely in browser RAM, the tool can seamlessly process tens of thousands of rows within seconds. For datasets exceeding 500,000+ rows, ensure sufficient available system RAM.